

EMINENCE 6.0

Artificial Intelligence-Based Task

By

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Time Allocation – 2 hours (for both AI Tasks)

Rules for Both Tasks

- **Use the starter notebook** and follow the rules written in it. Do not edit Section 8 or any cell marked "DO NOT MODIFY".
- **Use of AI tools is strictly prohibited**
- **Folder layout:** keep your training file in a data folder next to the notebook (inside the AI Task folder of your team repository). The organizers add the hidden evaluation file to the same folder.
- **Use only the provided training data.** External datasets, hardcoded labels and lookup logic lead to disqualification.
- **Approach summary:** write a short summary in your own words (5–8 lines) in the cell provided at the end of Section 7.
- Push your work to GitHub under the AI Task folder in your team repository.

Task 01 – Customer Churn Prediction (Classification)

Problem Statement

LankaTel, a telecommunications company wants to understand why customers stop using their services. Customer churn is when a subscriber cancels their plan and moves to a competitor. It is one of the most expensive problems in the telecom industry, since acquiring a new customer, costs far more than retaining an existing one.

The company has collected detailed records for thousands of customers, covering the services they use, how they are billed, and basic demographic details. The goal is to identify patterns that separate customers who are likely to leave from those who are likely to stay, so that the company can act early with targeted retention offers.

Your task is to **build a Machine Learning Classification Model** that can **predict whether a customer will churn based on these features**.

Dataset Information

The training dataset can be downloaded from the Google Drive link shared with your team. The hidden evaluation file has the same columns as the training file, except that customerID is removed.

Columns in the Dataset

The dataset contains account, service, and demographic information for each customer collected by LankaTel.

- customerID: Unique ID of each customer (training file only; do not use it as a feature)
- gender: Customer's gender (Male/Female)
- SeniorCitizen: Whether the customer is a senior citizen (0 = No, 1 = Yes)
- Partner: Whether the customer has a partner (Yes/No)
- Dependents: Whether the customer has dependents (Yes/No)
- tenure: Number of months the customer has stayed with the company
- PhoneService: Whether the customer has phone service (Yes/No)
- MultipleLines: Whether the customer has multiple lines (Yes/No/No phone service)
- InternetService: Customer's internet service provider (DSL/Fiber optic/No)
- OnlineSecurity: Whether the customer has online security add-on (Yes/No/No internet service)
- OnlineBackup: Whether the customer has online backup add-on (Yes/No/No internet service)
- DeviceProtection: Whether the customer has device protection add-on (Yes/No/No internet service)
- TechSupport: Whether the customer has tech support add-on (Yes/No/No internet service)
- StreamingTV: Whether the customer has streaming TV (Yes/No/No internet service)
- StreamingMovies: Whether the customer has streaming movies (Yes/No/No internet service)
- Contract: Contract term (Month-to-month/One year/Two year)
- PaperlessBilling: Whether the customer uses paperless billing (Yes/No)
- PaymentMethod: Payment method (Electronic check/Mailed check/Bank transfer (automatic)/Credit card (automatic))
- MonthlyCharges: Amount charged to the customer monthly
- TotalCharges: Total amount charged to the customer
- Churn: Target variable – whether the customer churned (stored as True/False in the CSV files; True = churned). Convert it to 1/0 for training.

What You Need to Do

Data Cleaning and Preparation

- Split the raw data into train, validation and test sets **first**, before fitting anything.
- Explore and clean the dataset – handle missing/null values and categorical variables. Cleaning and feature functions must not drop rows. Remove duplicates or outliers (if you want to) from the train split only, using the `remove_bad_training_rows()` function in the notebook.
- Perform feature engineering to transform the given columns into more useful features for the algorithm.
- Put encoding, scaling and the model into one scikit-learn Pipeline and fit it on the train split only.

Model Selection and Training

- Train different classification models using the train set.
- Perform hyperparameter tuning using the validation set on the different models.

Model Evaluation

- Test the trained model with the unused test set.

- Report training, validation and test performance using the following metrics: Accuracy, Precision, Recall, F1-score and ROC-AUC Score.

For competition scoring, we will use **only the F1-score** (of the churn class) on the hidden evaluation set for this task. (Higher F1 = better performance.)

`predict_churn()` must return 0/1 predictions and the probability of class 1 (churn).

What You Need to Submit

- Notebook: use the starter notebook and follow the rules mentioned there.
- Push your work to GitHub under the AI Task folder in the relevant team repository.
- File naming rule: rename your notebook as `TeamName_Task1_CustomerChurn.ipynb`

Important: If you do not follow the rules in the starter notebook, and naming conventions, your submission will be considered invalid and will not be evaluated.

Task 02 – Used Car Price Prediction (Regression)

Problem Statement

An online used-car marketplace wants to help sellers list their vehicles at a fair, competitive price and help buyers understand whether an asking price is reasonable. Pricing a used car correctly depends on many factors such as how old it is, how far it has been driven, whether it has been in an accident, and what kind of engine and fuel type it has.

The marketplace has collected listing data for thousands of used cars that were recently sold, including their brand, model, age, mileage, and mechanical specifications. The goal is to train a model that can estimate a fair selling price for a car based on its characteristics, helping both buyers and sellers make better decisions.

Your task is to **build a Machine Learning Regression Model** that can **predict the price of a used car** based on its features.

Dataset Information

The training dataset can be downloaded from the Google Drive link shared with your team. The hidden evaluation file has the same columns as the training file.

Columns in the Dataset

- `brand`: Manufacturer of the car (e.g., Toyota, Honda, Ford)
- `model`: Model name of the car
- `model_year`: Year the car was manufactured
- `mileage`: Total distance driven, in miles
- `fuel_type`: Type of fuel the car uses (e.g., Gasoline, Diesel, Electric, Hybrid)
- `engine`: Engine description (horsepower, cylinder count, displacement)
- `transmission`: Transmission type (e.g., Automatic, Manual)
- `ext_col`: Exterior colour of the car
- `int_col`: Interior colour of the car
- `accident`: Whether the car has a reported accident history (Yes/No)
- `clean_title`: Whether the car has a clean title (Yes/No)

- price: Target variable – selling price of the car in US dollars, as a plain number. It is in both the training file and the hidden evaluation file; Section 8 of the notebook separates it before calling your function.

What You Need to Do

Data Cleaning and Preparation

- Split the raw data into train, validation and test sets **first**, before fitting anything.
- Explore and clean the dataset – handle missing/null values and categorical variables. Cleaning and feature functions must not drop rows or use price. Remove duplicates, price outliers or obvious price errors (if you want to) from the train split only, using the `remove_bad_training_rows()` function in the notebook.
- Perform feature engineering to transform the given columns into more useful features for the algorithm.
- Put encoding, scaling and the model into one scikit-learn Pipeline and fit it on the train split only.

Model Selection and Training

- Train different regression models using the train set.
- Perform hyperparameter tuning using the validation set on the different models.

Model Evaluation

- Test the trained model with the unused test set.
- Report training, validation and test performance using the following metrics: RMSE (Root Mean Squared Error), MAE (Mean Absolute Error) and R^2 score.

For competition scoring, we will use **only MAE** on the hidden evaluation set for this task. (Lower MAE = better performance.)

`predict_price()` must return prices in dollars (the original units), even if you train on $\log(\text{price})$.

What You Need to Submit

- Notebook: use the starter notebook and follow the rules mentioned there.
- Push your work to GitHub under the AI Task folder in the relevant team repository.
- File naming rule: rename your notebook as `TeamName_Task2_CarPrice.ipynb`

Important: If you do not follow the rules in the starter notebook, and naming conventions, your submission will be considered invalid and will not be evaluated.